

KARAN KAILAS MENDHE

ELECTRICAL ENGINEERING UNDERGRADUATE | POWER ELECTRONICS | EMBEDDED SYSTEMS | INDUSTRIAL AUTOMATION

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PROFESSIONAL SUMMARY

Electrical Engineering undergraduate with a strong foundation in power electronics, embedded systems, industrial automation, and hardware prototyping. Experienced in developing real-world engineering projects involving ESP32, IoT, microcontrollers, and electronic systems. Passionate about designing practical engineering solutions through hands-on learning, innovation, and continuous skill development. A quick learner with strong problem-solving, teamwork, leadership, and communication skills, seeking opportunities to contribute and grow in the electrical engineering industry.

EDUCATION

Bachelor of Technology (B.Tech) Electrical Engineering Bajaj Institute of Technology, Wardha Expected Graduation: 2027 CGPA: 6.60	Higher Secondary Certificate (HSC) Insight Junior College, Jamtha Maharashtra State Board Year: 2022 Percentage: 87.67%	Secondary School Certificate (SSC) Kasturba Vidya Mandir, Sevagram Maharashtra State Board Year: 2020 Percentage: 85.20%
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TECHNICAL SKILLS

Electrical Engineering - Power Electronics - Circuit Design - Electrical Machines - Basic Power Systems - Industrial Automation	Embedded Systems - ESP32 - Arduino - Sensors & Actuators - Relay & Motor Control	Design & Simulation - AutoCAD Electrical - MATLAB - Simulink - LTspice - Tinkercad	Hardware - Hardware Prototyping - PCB Fundamentals - Electronic Testing - Soldering	Programming - C - C++ - Python - Arduino IDE
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PROFESSIONAL SKILLS

Problem Solving | Analytical Thinking | Teamwork | Leadership | Effective Communication | Time Management | Adaptability | Attention to Detail

PROJECTS

Dextra – Smart Hand Orthosis (SIH 2025 Grand Finalist)

- Designed and developed a wearable smart hand orthosis using ESP32 and servo motors for wrist and finger rehabilitation.
- Implemented a tendon-driven mechanism for controlled finger actuation and precise hardware integration.
- Developed embedded control logic for reliable and user-friendly operation.
- Contributed to prototype assembly, testing, validation, and optimization for lightweight construction and comfort.
- Presented the project at the Smart India Hackathon 2025 Grand Finale as part of Team Dexteriest.

Smart Grid & Microgrid Monitoring System

- Developed an IoT-based monitoring system for real-time analysis of solar grid and microgrid parameters.
- Integrated voltage, current, power, temperature, and irradiance sensors with relay modules.
- Designed a real-time monitoring and visualization system for performance analysis.
- Improved system reliability, safety, and energy management through intelligent monitoring.
- Presented the project at Smartathon 2.0 National Level Innovation Competition.

DC–DC High Boost Gain Converter

- Designed and fabricated a high-gain DC–DC boost converter for efficient voltage conversion.
- Performed hardware assembly, testing, and performance evaluation.
- Improved converter efficiency through practical design optimization.

INDUSTRIAL & TECHNICAL TRAINING	CERTIFICATIONS	AREAS OF INTEREST	HOBBIES
Industrial Training – Electrical Loco Shed, Nagpur - Gained practical exposure to locomotive electrical systems. - Learned about traction motors, battery banks, industrial maintenance, and electrical safety practices. - Observed maintenance procedures and industrial operations in a real-world environment.	- MATLAB Onramp – MathWorks - Simulink Onramp – MathWorks - Introduction to Power Electronics Coursera (University of Colorado Boulder) - One-Week Short-Term Training Program (STTP): Smart Grid Control – Renewable Energy & Vehicle Integration	- Power Electronics - Embedded Systems - Industrial Automation - Electrical Design - Renewable Energy - Electric Vehicles (EV)	- Building Engineering Projects - Learning Emerging Technologies - Reading Technical Articles - Exploring Renewable Energy & EV Technologies - Movies & Documentaries - Swimming

RELEVANT COURSEWORK

- Power Electronics | Electrical Machines | Control Systems | Power Systems | Industrial Automation | Embedded Systems | Analog & Digital Electronics

LANGUAGES - English (Professional Working Proficiency) - Hindi (Native/Bilingual Proficiency) - Marathi (Native/Bilingual Proficiency)	ACHIEVEMENTS - Smart India Hackathon 2025 – Grand Finalist (Team Dexteriest)(Medtech,Healthcare) - Smartathon 2.0 – National Level Finalist (Team Raynex)(Renewable Energy) - Wardha Innovation Challenge – Runner-up (Pitching , Market ready product)
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